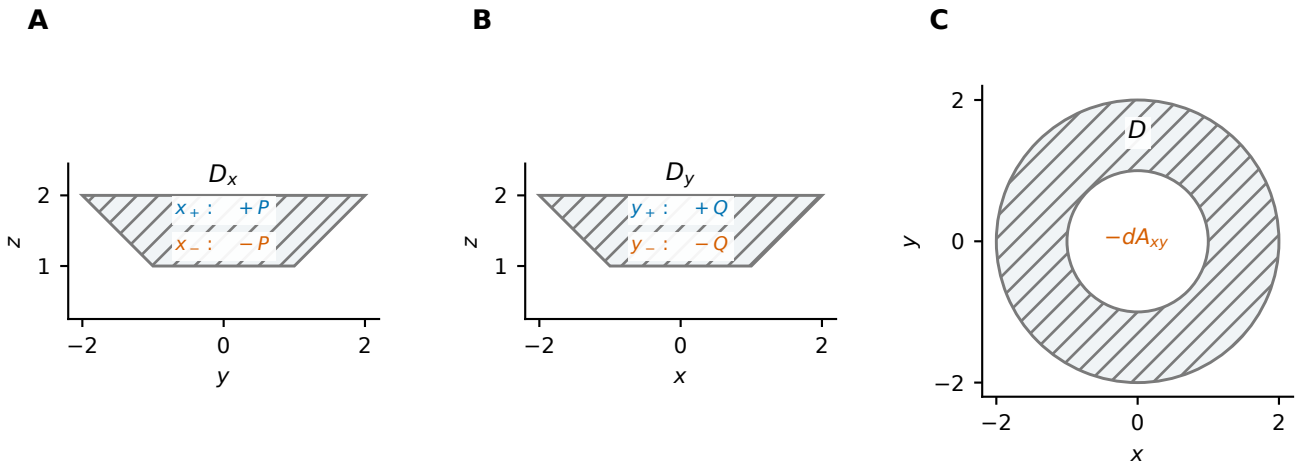

$dy\,dz = \frac{x}{r}\,dA_{xy}$

$dz\,dx = \frac{y}{r}\,dA_{xy}$

$dx\,dy = -dA_{xy}$